

**Bachelor of Technology — Electrical Engineering**

Table 1: Electrical Engineering — B24 Credit Distribution

| Division           | Subdivision                 | Credits    |
|--------------------|-----------------------------|------------|
| IC Core Courses    | IC Compulsory               | 39         |
|                    | IC Basket                   | 6          |
|                    | IKS + HSS                   | 15         |
| Discipline Courses | Discipline Core             | 52         |
|                    | Discipline Electives        | 20         |
| Electives          | Free Electives              | 17         |
|                    | MTP (4+4, equivalent as DE) | 8          |
|                    | ISTP (equivalent as FE)     | 4          |
| <b>Total</b>       |                             | <b>161</b> |

Table 2: B. Tech Electrical Engineering — B24 Discipline Core Courses

| Sl. No       | Course Code | Course Name                                             | L-T-P-C     | Credits   |
|--------------|-------------|---------------------------------------------------------|-------------|-----------|
| 1            | EE-203      | Network Theory                                          | 3-0-0-3     | 3         |
| 2            | EE-210      | Digital Systems Design                                  | 3-0-0-3     | 3         |
| 3            | EE-210P     | Digital Systems Design Lab                              | 0-0-2-1     | 1         |
| 4            | EE-260      | Signals and Systems                                     | 2.5-0.5-0-3 | 3         |
| 5            | EE-261      | Electrical Systems Around Us                            | 3-0-0-3     | 3         |
| 6            | EE-261P     | Electrical Systems Around Us Lab                        | 0-0-3-2     | 2         |
| 7            | EE-311      | Device Electronics                                      | 3-0-0-3     | 3         |
| 8            | EE-201      | Electromechanics                                        | 2.5-0.5-0-3 | 3         |
| 9            | EE-201P     | Electromechanics Lab                                    | 0-0-2-1     | 1         |
| 10           | EE-205      | Electromagnetics & Wave Propagation                     | 3-0-0-3     | 3         |
| 11           | EE-211      | Analog Circuit Design                                   | 2-0-2-3     | 3         |
| 12           | EE-223P     | Reverse Engineering                                     | 1-0-0-1     | 1         |
| 13           | EE-304      | Communication Systems                                   | 3-0-1-4     | 4         |
| 14           | EE-231      | Measurement and Instrumentation                         | 2-0-2-3     | 3         |
| 15           | EE-302      | Control Systems                                         | 3-0-2-4     | 4         |
| 16           | EE-303      | Power System                                            | 3-1-0-4     | 4         |
| 17           | EE-314      | Digital Signal Processing                               | 3-0-2-4     | 4         |
| 18           | EE-326      | Computer Organization and Processor Architecture Design | 3-0-2-4     | 4         |
| <b>Total</b> |             |                                                         |             | <b>52</b> |

Table 3: B. Tech Electrical Engineering — B24 Semester 1

| Sl. No | Course Category         | Course Code | Course Name                                      | L-T-P-C     | Credits |
|--------|-------------------------|-------------|--------------------------------------------------|-------------|---------|
| 1      | IC                      | IC-112      | Calculus                                         | 1.5-0.5-0-2 | 2       |
| 2      | IC                      | IC-113      | Complex Variables and Vector Calculus            | 1.5-0.5-0-2 | 2       |
| 3      | IC                      | IC-140      | Engineering Graphics for Design                  | 2-0-3-4     | 4       |
| 4      | IC                      | IC-152      | Introduction to Python and Data Science          | 3-0-2-4     | 4       |
| 5      | IC Basket<br>(choose 1) | IC-131      | Applied Chemistry for Engineers                  | 2-0-2-3     | 3       |
| 6      |                         | IC-136      | Understanding Biotechnology and its Applications | 3-0-0-3     |         |
| 7      |                         | IC-230      | Environmental Science                            | 3-0-0-3     |         |
| 8      | IKS + HSS<br>(choose 1) | HS-108      | Basic English for Engineers                      | 3-0-0-3     | 3       |
| 9      |                         | HS-110      | Japanese for Beginners                           | 3           |         |
| 10     |                         | HS-151      | Introduction to English Literature               | 3-0-0-3     |         |
| 11     |                         | HS-256      | Introduction to Spanish                          | 3           |         |
| Total  |                         |             |                                                  |             | 18      |

*Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.*

Table 4: B. Tech Electrical Engineering — B24 Semester 2

| Sl. No       | Course Category         | Course Code | Course Name                       | L-T-P-C     | Credits   |
|--------------|-------------------------|-------------|-----------------------------------|-------------|-----------|
| 1            | IC                      | IC-102P     | Foundations of Design Practicum   | 1-0-6-4     | 4         |
| 2            | IC                      | IC-114      | Linear Algebra                    | 2-0-0-2     | 2         |
| 3            | IC                      | IC-115      | ODE & Integral Transforms         | 2-0-0-2     | 2         |
| 4            | IC                      | IC-161      | Applied Electronics               | 3           | 3         |
| 5            | IC                      | IC-161P     | Applied Electronics Lab           | 2           | 2         |
| 6            | IC                      | IC-252      | Probability and Statistics        | 3-0-2-4     | 4         |
| 7            | IC Basket<br>(choose 1) | IC-121      | Mechanics of Particles and Waves  | 2.5-0.5-0-3 | 3         |
| 8            |                         | IC-240      | Mechanics of Rigid Bodies         | 1.5-1.5-0-3 |           |
| 9            |                         | IC-241      | Material Science for Engineers    | 3-0-0-3     |           |
| 10           |                         | IC-253      | Data Structures and Algorithms    | 2-0-2-3     |           |
| 11           | IKS + HSS               | IC-182      | History of Science and Technology | 3           | 3         |
| <b>Total</b> |                         |             |                                   |             | <b>23</b> |

*Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.*

Table 5: B. Tech Electrical Engineering — B24 Semester 3

| Sl. No       | Course Category | Course Code | Course Name                      | L-T-P-C     | Credits   |
|--------------|-----------------|-------------|----------------------------------|-------------|-----------|
| 1            | IC              | IC-202P     | Design Practicum                 | 0-0-6-3     | 3         |
| 2            | IC              | IC-272      | Machine Learning                 | 2-0-2-3     | 3         |
| 3            | DC              | EE-203      | Network Theory                   | 3-0-0-3     | 3         |
| 4            | DC              | EE-210      | Digital Systems Design           | 3-0-0-3     | 3         |
| 5            | DC              | EE-210P     | Digital Systems Design Lab       | 0-0-2-1     | 1         |
| 6            | DC              | EE-260      | Signals and Systems              | 2.5-0.5-0-3 | 3         |
| 7            | DC              | EE-261      | Electrical Systems Around Us     | 3-0-0-3     | 3         |
| 8            | DC              | EE-261P     | Electrical Systems Around Us Lab | 0-0-3-2     | 2         |
| 9            | DC              | EE-311      | Device Electronics               | 3-0-0-3     | 3         |
| <b>Total</b> |                 |             |                                  |             | <b>24</b> |

*Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.*

Table 6: B. Tech Electrical Engineering — B24 Semester 4

| Sl. No       | Course Category | Course Code | Course Name                         | L-T-P-C     | Credits   |
|--------------|-----------------|-------------|-------------------------------------|-------------|-----------|
| 1            | IC              | IC-222P     | Physics Practicum                   | 0-0-3-2     | 2         |
| 2            | DC              | EE-201      | Electromechanics                    | 2.5-0.5-0-3 | 3         |
| 3            | DC              | EE-201P     | Electromechanics Lab                | 0-0-2-1     | 1         |
| 4            | DC              | EE-205      | Electromagnetics & Wave Propagation | 3-0-0-3     | 3         |
| 5            | DC              | EE-211      | Analog Circuit Design               | 2-0-2-3     | 3         |
| 6            | DC              | EE-223P     | Reverse Engineering                 | 1-0-0-1     | 1         |
| 7            | DC              | EE-304      | Communication Systems               | 3-0-1-4     | 4         |
| <b>Total</b> |                 |             |                                     |             | <b>17</b> |

*Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.*

Table 7: B. Tech Electrical Engineering — B24 Semester 5

| Sl. No       | Course Category | Course Code | Course Name                                             | L-T-P-C | Credits   |
|--------------|-----------------|-------------|---------------------------------------------------------|---------|-----------|
| 1            | DC              | EE-231      | Measurement and Instrumentation                         | 2-0-2-3 | 3         |
| 2            | DC              | EE-302      | Control Systems                                         | 3-0-2-4 | 4         |
| 3            | DC              | EE-303      | Power System                                            | 3-1-0-4 | 4         |
| 4            | DC              | EE-314      | Digital Signal Processing                               | 3-0-2-4 | 4         |
| 5            | DC              | EE-326      | Computer Organization and Processor Architecture Design | 3-0-2-4 | 4         |
| <b>Total</b> |                 |             |                                                         |         | <b>19</b> |

*Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.*

Table 8: B. Tech Electrical Engineering — B24 Semester 6

| Sl. No       | Course Category | Course Code | Course Name                                 | L-T-P-C | Credits  |
|--------------|-----------------|-------------|---------------------------------------------|---------|----------|
| 1            | ISTP            | DP-301P     | Interdisciplinary Socio-Technical Practicum | 0-0-6-4 | 4        |
| <b>Total</b> |                 |             |                                             |         | <b>4</b> |

*Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.*

Table 9: B. Tech Electrical Engineering — B24 Semester 7

| Sl. No       | Course Category | Course Code | Course Name                 | L-T-P-C | Credits  |
|--------------|-----------------|-------------|-----------------------------|---------|----------|
| 1            | MTP             | EE-498P     | Major Technical Project - I | 0-0-6-4 | 4        |
| <b>Total</b> |                 |             |                             |         | <b>4</b> |

*Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.*

Table 10: B. Tech Electrical Engineering — B24 Semester 8

| Sl. No       | Course Category | Course Code | Course Name                  | L-T-P-C | Credits  |
|--------------|-----------------|-------------|------------------------------|---------|----------|
| 1            | IC              | EE-010      | Internship                   |         | 2        |
| 2            | MTP             | EE-499P     | Major Technical Project - II | 0-0-6-4 | 4        |
| <b>Total</b> |                 |             |                              |         | <b>6</b> |

*Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.*

*Beyond the core courses above, students choose 20 credits of **Discipline Electives** and 17 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 11: Electrical Engineering — Available Discipline Electives (choose from these)

| Sl. No | Course Code | Course Name                               | Credits |
|--------|-------------|-------------------------------------------|---------|
| 1      | BE-302      | Bioelectric Systems Modeling              | 4       |
| 2      | BE-304      | Bioinformatics                            | 4       |
| 3      | CS-549      | Performance Analysis of Computer Networks | 3       |
| 4      | CS-671      | Deep Learning and Applications            | 4       |
| 5      | EE-500      | Network Control System                    | 3       |
| 6      | EE-501      | Power System Operation and Control        | 3       |
| 7      | EE-502      | Analog System Design Laboratory           | 3       |
| 8      | EE-503      | Advance Communication Theory              | 3       |
| 9      | EE-504      | Switch Mode Power Conversion              | 3       |
| 10     | EE-505      | Materials for Semiconductor Industry      | 3       |
| 11     | EE-506      | Solar Photovoltaic Energy Systems         | 3       |

*Electrical Engineering — Available Discipline Electives (choose from these) (continued)*

| Sl. No | Course Code | Course Name                                                     | Credits |
|--------|-------------|-----------------------------------------------------------------|---------|
| 12     | EE-507      | Transmission Lines and Basic Microwave Engineering              | 4       |
| 13     | EE-508      | Fundamentals of Electric Drives                                 | 3       |
| 14     | EE-508P     | Practicum on fundamentals of Electrical Drives                  | 2       |
| 15     | EE-509      | Linear Dynamical Systems                                        | 3       |
| 16     | EE-511      | Computer Vision                                                 | 4       |
| 17     | EE-512      | CMOS Analog IC Design                                           | 4       |
| 18     | EE-513      | Special Electrical Machines                                     | 3       |
| 19     | EE-514      | Robust Control Systems                                          | 3       |
| 20     | EE-515      | Nonlinear Stability and Control                                 | 4       |
| 21     | EE-516      | Biomedical Systems                                              | 4       |
| 22     | EE-517      | Wireless Communications and Networking                          | 3       |
| 23     | EE-518      | Information Theory                                              | 3       |
| 24     | EE-519      | CMOS Digital IC design Practicum                                | 2       |
| 25     | EE-519P     | CMOS Digital IC Design Practicum                                | 2       |
| 26     | EE-520      | Microelectronic Devices and Modelling                           | 3       |
| 27     | EE-522      | Matrix Theory for Engineers                                     | 3       |
| 28     | EE-523      | Digital VLSI Architecture Design                                | 4       |
| 29     | EE-524      | Digital MOS LSI Circuits                                        | 3       |
| 30     | EE-526      | Power Semiconductor Devices                                     | 3       |
| 31     | EE-527      | Analysis and Design of Power Electronic Converters              | 3       |
| 32     | EE-527P     | Practicum on Analysis and Design of Power Electronic Converters | 2       |
| 33     | EE-528      | Modeling and Analysis of Electric Machines                      | 3       |
| 34     | EE-529      | Embedded Systems                                                | 4       |
| 35     | EE-530      | Optimization theory                                             | 3       |
| 36     | EE-531      | Estimation and Detection Theory                                 | 3       |
| 37     | EE-532      | Supervised Research Exposure                                    | 3       |
| 38     | EE-533      | Access Network                                                  | 4       |
| 39     | EE-534      | Probability and Random Processes                                | 3       |
| 40     | EE-535      | Communication and Signal Processing Systems Design              | 3       |
| 41     | EE-536      | IoT Systems                                                     | 3       |
| 42     | EE-537      | Power Quality Problems and Mitigation Techniques in Microgrids  | 3       |
| 43     | EE-538      | Data-driven Control                                             | 3       |
| 44     | EE-540      | Wide Band Gap Devices in Power Electronics Applications         | 3       |
| 45     | EE-541      | Sensors: Techniques, Algorithms, Applications for Signal        | 4       |
| 46     | EE-542      | Modelling, Simulation and Control of Hybrid Electric Vehicle    | 3       |
| 47     | EE-543      | Vision and Learning Based Control                               | 3       |

*Electrical Engineering — Available Discipline Electives (choose from these) (continued)*

| Sl. No | Course Code | Course Name                                                     | Credits |
|--------|-------------|-----------------------------------------------------------------|---------|
| 48     | EE-550      | Optimal Control and Estimation                                  | 3       |
| 49     | EE-551      | Applied Photonics for Scientists and Engineers                  | 3       |
| 50     | EE-552      | Power and Energy Systems                                        | 4       |
| 51     | EE-553      | Foundations of Intelligent Communication Systems-I              | 4       |
| 52     | EE-554      | Low Power VLSI Design                                           | 3       |
| 53     | EE-555      | Intelligent Control System                                      | 3       |
| 54     | EE-556      | Nuclear Reactor Control                                         | 1       |
| 55     | EE-557      | Adaptive Control                                                | 3       |
| 56     | EE-560      | Reconfigurable Computing                                        | 4       |
| 57     | EE-570      | Advanced Communication Theory                                   | 4       |
| 58     | EE-574      | Biomedical Signal and Image Analysis                            | 3       |
| 59     | EE-575      | Applied Statistics for Data and Signals                         | 4       |
| 60     | EE-580      | Network Systems: Modelling and Analysis                         | 3       |
| 61     | EE-581      | Applied Statistics for Data and Signals                         | 4       |
| 62     | EE-582      | Nonlinear Analysis and Control of Power Electronic Converters   | 3       |
| 63     | EE-583      | Smart Grids                                                     | 3       |
| 64     | EE-584      | Power System Protection                                         | 3       |
| 65     | EE-585      | Theory, Modelling and Characterization of Photo-voltaic Devices | 3       |
| 66     | EE-593      | Low Power VLSI Circuits                                         | 3       |
| 67     | EE-597      | Intellegent Control System                                      | 3       |
| 68     | EE-601      | Solid State Devices                                             | 3       |
| 69     | EE-602      | Control System Applications                                     | 3       |
| 70     | EE-603      | Renewable Energy and Smart Grid                                 | 3       |
| 71     | EE-604      | Practicum on Advanced Electric Drives                           | 3       |
| 72     | EE-604P     | Practicum on Advanced Electric Drives                           | 1       |
| 73     | EE-605      | Information Theory                                              | 3       |
| 74     | EE-606      | Introduction to High Voltage Engineering and Dielectric         | 3       |
| 75     | EE-607      | Optical Communication Systems                                   | 3       |
| 76     | EE-608      | Digital Image Processing                                        | 4       |
| 77     | EE-609      | Network Information Theory                                      | 3       |
| 78     | EE-611      | VLSI Technology                                                 | 3       |
| 79     | EE-611P     | VLSI Fabrication Practicum                                      | 2       |
| 80     | EE-612      | OFDM For Optical Communications                                 | 3       |
| 81     | EE-613      | Wireless Communication                                          | 3       |
| 82     | EE-614      | Optical Communication Systems                                   | 3       |
| 83     | EE-615      | Nanoelectronics and Micronanofabrication                        | 3       |
| 84     | EE-616      | Microwave engineering                                           | 2       |
| 85     | EE-618      | Industrial Process Control                                      | 3       |

*Electrical Engineering — Available Discipline Electives (choose from these) (continued)*

| Sl. No                                                               | Course Code | Course Name                                                  | Credits |
|----------------------------------------------------------------------|-------------|--------------------------------------------------------------|---------|
| 86                                                                   | EE-619      | Mixed Signal VLSI Design                                     | 4       |
| 87                                                                   | EE-620      | Advanced Digital Signal Processing                           | 3       |
| 88                                                                   | EE-621      | Radiating systems                                            | 4       |
| 89                                                                   | EE-622      | Microwave Integrated Circuits                                | 3       |
| 90                                                                   | EE-623P     | Practicum on Digital Control of Power Electronics and Drives | 3       |
| 91                                                                   | EE-626P     | Post Graduate Project - I                                    | 10      |
| 92                                                                   | EE-630      | HVDC Transmission and Flexible AC Transmission Systems       | 3       |
| 93                                                                   | EE-641      | Advance Wireless Technologies                                | 4       |
| 94                                                                   | EE-642      | Research Study                                               | 3       |
| 95                                                                   | EE-677      | Analog Circuit Design                                        | 3       |
| 96                                                                   | EE-678      | Digital Circuit Design                                       | 3       |
| 97                                                                   | EE-691      | Advanced Topics in Dielectric Breakdown                      | 1       |
| 98                                                                   | EE-694      | Photonics Computing                                          | 3       |
| <i>Compulsory project components — counted as DE, not selectable</i> |             |                                                              |         |
| 99                                                                   | EE-498P     | Major Technical Project - I (compulsory; counted as DE)      | 4       |
| 100                                                                  | EE-499P     | Major Technical Project - II (compulsory; counted as DE)     | 4       |